

Sharp Radical Bounds and Survivor-First Compression in Exact Unknot Recognition

Finite homological perturbation, certified minimal models,
and a parameterized route to quasi-polynomial complexity

Research contribution prepared for the ProveIt project

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Abstract

We develop a chain-level complement to the earlier twist and component-compression work in ProveIt’s unknot-recognition program. For the characteristic-two arc category on $2k$ boundary points, we give an explicit proof of the sharp radical bound $J^{2k} = 0 \neq J^{2k-1}$ for $k \geq 1$. The proof uses a dot derivative and a nested-matching witness. A central marked-dot splitting further identifies the algebra as $B \otimes \mathbb{F}_2[z]/(z^2)$, with radical index $2k - 1$ for B .

For any bounded complex in this category, the scalar residue predicts the exact multiplicity of every matching in every degree of a minimal representative: $\mu_{a,h} = m_{a,h} - r_{a,h} - r_{a,h-1}$. This yields both a cheap survivor oracle and a proof that changing pivot order cannot improve these final multiplicities for a fixed processed tangle. A finite perturbation formula constructs the remaining differential directly on the predicted survivors. Its maximum necessary correction order is $2k - 2$, and we exhibit complexes attaining that order. The expensive composition count is output-sensitive in the survivor count, subject to explicit input-validation and certificate-cost qualifications.

The package contains exact standard-library Python implementations, full homotopy-identity checks, an independently checked braid-scanning harness, and paired benchmarks. It checks 306 braid closures against a crossing-cube oracle and verifies 1,365 intermediate-stage contractions. On one alternating three-braid benchmark, the local transfer scanner is about 2.23 times faster than the included scalar-pivot baseline; other cases are slower. These are not benchmarks against the optimized upstream recognizer. The resulting complexity theorem is unconditional in boundary width and minimal survivor count, but no uniform quasi-polynomial bound for arbitrary knot diagrams is established.

Status. Complete written proofs of the algebraic and parameterized statements; executable exact-arithmetic checks; no Lean formalization, no first-in-literature priority claim, and no claim to have run the full upstream repository test suite.

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1 Research objective, existing work, and precise scope

The objective is an exact, practically useful unknot recognizer with an eventual worst-case running time $n^{O(\log n)}$, where n measures an explicit diagram. There are two distinct questions. One asks how to reduce the cost of the exact Khovanov pipeline that already exists. The other asks for a global structural argument forcing a quasi-polynomial bound. A faster numerical kernel, a smaller intermediate complex on examples, and a global complexity theorem are different achievements. This report treats them separately.

The inspected ProveIt subtree documents an exact but worst-case exponential implementation. Its scanner already includes bit-packed dotted cobordisms, compiled gluing plans, isomorphism cancellation, fill-sensitive pivot selection, and alternative scan orders [8]. The earlier project contribution on twist compression replaces long local crossing blocks by smaller complexes; the component-compression contribution accelerates individual cobordism compositions and studies hierarchy restart counts [10, 11]. We do not reclassify those existing improvements as results of this report.

The present work addresses a different bottleneck: repeatedly performing cobordism-valued Schur updates before knowing how many objects will survive. It separates a differential into a readily processed scalar part and a nilpotent remainder. The scalar part determines the final minimal object counts exactly. The remainder can then be transferred to those objects in a uniformly bounded number of algebraic correction stages.

1.1 Connection with the two referenced repositories

The methodological connection to `openai/math` is its finite Artin-model construction in the section entitled “Finite cochain models” of the binary algebra manuscript [9]. There, residual cohomology is lifted to a finite model using a nilpotent perturbation. We use the same general strategy, but none of that manuscript’s arithmetic conclusions is an input to our proofs. The homological perturbation lemma itself is classical; an appropriate primary reference is Crainic [4]. Our category has several nonisomorphic matching objects, rather than a single local coefficient ring, and its exact characteristic-two radical index supplies a concrete transfer bound.

The original arc algebra and tangle formalism are due to Khovanov and Bar-Natan [1, 2, 3]. Minimal complexes over a nilpotent radical and the perturbation lemma are established algebraic tools. Our contribution is an explicit, independently proved specialization, its sharpness witnesses, its output-sensitive implementation, and its integration with the particular recognition pipeline. We have not established bibliographical priority for the sharp nilpotence statement or the marked splitting.

1.2 The two obstructions that remain

Even when the boundary is small, the number of copies of its matching objects can be large. This distinction is essential. Kelomäki–Schütz prove exponential behavior for basis-expanding scanning on certain positive and alternating three-braid families, while also giving a different polynomial-time algorithm for three-braid Khovanov homology [6]. Thus bounded strand number alone cannot justify a polynomial bound on an explicitly expanded minimal complex. Our survivor theorem makes this multiplicity obstruction visible before the expensive elimination phase.

The separate hierarchy route also remains relevant. Lackenby’s 2026 preprint explains in Section 9

that its algorithmic steps have not been specified precisely enough to estimate their running times [7]. This report neither supplies those missing geometric routines nor derives their complexity bounds from Khovanov calculations. Its quasi-polynomial consequence instead has an explicit boundary-and-survivor hypothesis.

2 The characteristic-two arc category and its intrinsic weight

Fix $2k$ labeled points in a prescribed cyclic order. Let $\mathcal{B}_k$ be their set of noncrossing perfect matchings. Labels carry incidence information; changing their names does not change the algebra. For $a, b \in \mathcal{B}_k$, let $c(a, b)$ be the number of circles obtained by overlaying a with the reflected matching b . Each circle alternates between arcs of the two matchings, with a coincident pair counted as a two-edge circle. In particular,

$$1 \leq c(a, b) \leq k \quad (k > 0), \quad c(a, b) = k \iff a = b.$$

We work over $\mathbb{F}_2$. After delooping, the dotted-cobordism morphism space is

$$\mathrm{Hom}(a, b) = \mathbb{F}_2[x_1, \dots, x_{c(a,b)}] / (x_1^2, \dots, x_{c(a,b)}^2) \quad (1)$$

as a vector space, with its indicated monomial basis. For unequal endpoints this notation specifies the basis, not a multiplication internal to that Hom space. Composition is the usual saddle-gluing operation. The Frobenius algebra on a circle is

$$A = \mathbb{F}_2[x]/(x^2), \quad \Delta(1) = 1 \otimes x + x \otimes 1, \quad \Delta(x) = x \otimes x. \quad (2)$$

A sphere without a dot is zero, a sphere with one dot is one, and positive-genus connected components evaluate to zero in this characteristic. These are precisely the numerical conventions used by the inspected composition routines.

The additive category of finite direct sums of these matchings is denoted $\mathcal{C}_k$. Equivalently, use the finite-dimensional algebra

$$H_k = \bigoplus_{a,b \in \mathcal{B}_k} \mathrm{Hom}(a, b),$$

with products zero when endpoints do not match. Our convention for gf is “ g after f .” Quantum shifts are forgotten, while homological degrees are retained. This is enough to compute total Khovanov ranks. The intrinsic weight below is an algebra grading; it is not the homological degree and is not a claim that all entries of an unshifted differential have the same weight.

Definition 2.1 (Intrinsic weight). For a basis monomial $x_S : a \rightarrow b$, define

$$\mathrm{wt}(x_S) = k - c(a, b) + 2|S|. \quad (3)$$

Let J be the span of all positive-weight basis monomials in H_k .

Lemma 2.2 (Additivity of weight). *Every nonzero monomial in the product of homogeneous morphisms of weights u and v has weight $u + v$. All weights lie between 0 and $2k$. The weight-zero space consists exactly of the scalar identities of the individual matchings.*

Proof. Consider basis inputs in $\mathrm{Hom}(a, b)$ and $\mathrm{Hom}(b, c)$, and put $p = c(a, b)$, $q = c(b, c)$, and $r = c(a, c)$. Glue the $p + q$ input discs along the k middle arcs. If a component has positive genus,

its contribution is zero, so suppose all surviving components have genus zero. Let t be their number. The Euler characteristic gives

$$p + q - k = 2t - r.$$

A genus-zero component with s output boundaries contributes $s - 1$ output dots when it has no input dot, and s output dots when it has one input dot. Two or more dots on that component give zero. Consequently a surviving output has $r - t + d$ dots, where d is the total input dot count. Its weight is

$$k - r + 2(r - t + d) = 2k - p - q + 2d,$$

which is the sum of the input weights. In these arc compositions every component has output boundary; the same Euler accounting also accommodates any scalar sphere evaluation in a larger dotted-cobordism presentation.

Nonnegativity follows from $c(a, b) \leq k$. The largest possible weight in a Hom space is $k + c(a, b) \leq 2k$. Weight zero forces $c(a, b) = k$ and no dots, hence $a = b$ and the identity basis element. $\square$

Corollary 2.3 (Residue algebra). *The space J is a nilpotent two-sided ideal, initially with the bound $J^{2k+1} = 0$, and*

$$H_k/J \cong \prod_{a \in \mathcal{B}_k} \mathbb{F}_2. \quad (4)$$

It is the Jacobson radical of H_k .

Proof. The ideal and nilpotence assertions follow from weight additivity. The quotient is the weight-zero algebra. A nilpotent ideal is contained in the Jacobson radical, since $1 - u$ has a finite geometric inverse for every element of that ideal, even after multiplication by an arbitrary algebra element. Conversely, a semisimple quotient has zero radical. Applying these two observations proves the final assertion. $\square$

The projection in [equation \(4\)](#) is especially inexpensive: discard every entry between distinct matchings, and retain only the constant coefficient of an entry between equal matchings. In the repository's packed encoding this coefficient is the low bit. The nontrivial fact is that discarding the other entries respects composition; [theorem 2.2](#) proves exactly this fact.

3 Dot differentiation and the sharp radical index

3.1 A derivation in characteristic two

For a Hom-space monomial, define

$$\partial(x_S) = \sum_{i \in S} x_{S \setminus \{i\}}. \quad (5)$$

This definition is independent of the numbering of the overlay circles. It lowers weight by two and satisfies $\partial^2 = 0$.

Lemma 3.1 (Leibniz rule for arc composition). *For composable morphisms f, g ,*

$$\partial(gf) = (\partial g)f + g(\partial f). \quad (6)$$

Proof. On A , write $D(1) = 0$ and $D(x) = 1$. Multiplication obeys $D\mu = \mu(D \otimes 1 + 1 \otimes D)$. The only possibly misleading case is $x \otimes x$: the right side is $x + x = 0$, as required. Similarly,

$$(D \otimes 1 + 1 \otimes D)\Delta(1) = 0 = \Delta D(1),$$

whereas its value on $\Delta(x)$ is $1 \otimes x + x \otimes 1 = \Delta D(x)$. Thus both elementary saddles commute with the total dot derivative on all their circle factors. Arc composition is a succession of these saddles, so the identity persists under composition. Before the saddles, the total derivative on the tensor product of the two input spaces is precisely the right side of [equation \(6\)](#). No assertion of derivative naturality for arbitrary cups or caps is needed here. $\square$

Theorem 3.2 (Sharp characteristic-two radical bound). *For every $k \geq 1$,*

$$J^{2k} = 0 \quad \text{and} \quad J^{2k-1} \neq 0. \quad (7)$$

The nilpotence index is therefore exactly $2k$. For $k = 0$, $H_0 = \mathbb{F}_2$ and $J = 0$.

Upper bound. It is enough to consider products of homogeneous radical basis elements. A product of $2k$ such factors can survive the weight bound only if every factor has weight one and the output has weight $2k$. A weight-one monomial is undotted, so its derivative is zero. By [theorem 3.1](#), the product has zero derivative.

Weight $2k$ occurs only in a diagonal Hom space, where it is spanned by the fully dotted monomial $x_1 \cdots x_k$. Its derivative is

$$\sum_{i=1}^k \prod_{j \neq i} x_j \neq 0.$$

Distinct summands are distinct basis monomials, including the single constant summand when $k = 1$. Thus a nonzero weight- $2k$ output cannot have zero derivative. The proposed product vanishes. Linearity proves $J^{2k} = 0$. $\square$

Explicit sharpness witness. Number the endpoints $0, 1, \dots, 2k - 1$. For $0 \leq j \leq k - 1$, set

$$a_j = \{(0, 2j + 1)\} \cup \{(2s - 1, 2s) : 1 \leq s \leq j\} \cup \{(2s, 2s + 1) : j + 1 \leq s \leq k - 1\}. \quad (8)$$

These are noncrossing matchings. Consecutive matchings differ by one saddle, and the undotted morphism between them has weight one. Follow the path

$$a_0 \longrightarrow a_1 \longrightarrow \cdots \longrightarrow a_{k-1} \longrightarrow a_{k-2} \longrightarrow \cdots \longrightarrow a_0$$

using undotted morphisms throughout. There are $2k - 2$ factors.

To evaluate the round trip, first note that the forward path merges one further circle at each step in $\text{Hom}(a_0, a_j)$. Its value is the undotted basis element, since $\mu(1 \otimes 1) = 1$. Reflection gives the analogous value for the reverse path. In their composite on the active $2j + 2$ endpoints, the two input discs are glued along $j + 1$ arcs and have $j + 1$ output circles. Its Euler characteristic is $1 - j$, hence its genus is zero. The resulting value is $\Delta^{(j)}(1)$, namely the sum of products of all but one of the $j + 1$ output variables. At $j = k - 1$, the complete round trip is therefore

$$\sum_{i=1}^k \prod_{j \neq i} x_j = \partial(x_1 \cdots x_k). \quad (9)$$

Multiplication by the endomorphism x_1 kills all but one of these summands and produces $x_1 \cdots x_k \neq 0$. This is a product of $2k - 1$ radical factors. For $k = 1$, the round trip has no factors and the final dot itself is the witness. $\square$

Remark 3.3 (Why the field matters). The argument uses $x + x = 0$ in the saddle identities. A proof over another field must not silently reuse it. The elementary weight bound $2k + 1$ is more general, but the improvement by one here is a characteristic-two assertion.

3.2 A central marked-dot splitting

Choose a boundary point p . For each matching, dot its arc incident to p ; the sum of these endomorphisms is denoted z . Sliding that dot across any morphism shows that z is central. Also $z^2 = 0$ and $\partial z = 1$. Set $B_k = \ker \partial \subset H_k$.

Proposition 3.4 (Splitting off one square-zero variable). *For $k \geq 1$, B_k is a subalgebra and multiplication gives an algebra isomorphism*

$$B_k \otimes_{\mathbb{F}_2} \mathbb{F}_2[z]/(z^2) \cong H_k. \quad (10)$$

For any morphism f , its unique decomposition is

$$f = f_0 + zf_1, \quad f_1 = \partial f, \quad f_0 = f + z\partial f, \quad f_0, f_1 \in B_k. \quad (11)$$

Moreover, if $J_B = \text{rad } B_k$, then

$$J^r = J_B^r \oplus zJ_B^{r-1} \quad (r \geq 1), \quad J_B^{2k-1} = 0 \neq J_B^{2k-2}. \quad (12)$$

For $k = 1$, the last inequality reads $B_1 \neq 0$ and $J_B = 0$.

Proof. The Leibniz rule makes B_k a subalgebra. Since $\partial(zf) = f + z\partial f$, both terms in [equation \(11\)](#) belong to B_k . This identity proves existence of the splitting. If $b_0 + zb_1 = 0$ with $b_i \in B_k$, applying ∂ gives $b_1 = 0$ and hence $b_0 = 0$. Centrality and $z^2 = 0$ now prove the algebra isomorphism.

The subalgebra B_k is graded. It has no nonzero elements of weight $2k$ by the fully dotted diagonal calculation. Weight $2k - 1$ can only occur when $c(a, b) = k - 1$ and every overlay circle is dotted; for $k \geq 2$ its derivative is again nonzero. When $k = 1$, no weight-one space exists. Thus B_k has maximum weight $2k - 2$. Its positive-weight ideal is its radical by the same argument as [theorem 2.3](#), so $J_B^{2k-1} = 0$. The undotted round trip in [equation \(9\)](#) belongs to J_B^{2k-2} and is nonzero, proving sharpness. Finally $J = J_B \oplus zB_k$, and an r -fold product contains at most one z . Expanding products as subspaces gives [equation \(12\)](#). $\square$

In each individual Hom space, the splitting halves the coefficient dimension. If the marked overlay circle is numbered zero, one may use $y_i = x_i + x_0$ for the other circles as a basis of $\ker \partial$; these variables also square to zero. This is a useful representation option, not a claim that the current Python code has already realized a factor-two memory saving. Its marked variant deliberately retains the original full encoding for independent identity checking.

4 The scalar residue predicts every minimal multiplicity

Let (C, d) be a bounded complex in $\mathcal{C}_k$. Write

$$C^h = \bigoplus_{a \in B_k} a^{\oplus m_{a,h}}.$$

The scalar residue $\bar{d}$ is obtained by applying [equation \(4\)](#) entrywise. It is a complex of finite-dimensional vector spaces, separately for each matching. Let $r_{a,h}$ be the binary rank of $\bar{d}_a^h : \mathbb{F}_2^{m_{a,h}} \rightarrow \mathbb{F}_2^{m_{a,h+1}}$.

Definition 4.1 (Minimal complex). A complex in $\mathcal{C}_k$ is minimal if every differential entry belongs to J . Equivalently, its scalar residue differential is zero. This is the same stopping condition as exhausting all invertible entries between equal matchings.

The equivalence follows because $\text{End}(a) = \mathbb{F}_2[x_1, \dots, x_k]/(x_i^2)$ is local: an element is a unit exactly when its constant coefficient is one. Distinct matching objects cannot be isomorphic, since their residues occupy different simple factors.

Theorem 4.2 (Exact survivor profile). *Every minimal complex homotopy equivalent to (C, d) has exactly*

$$\mu_{a,h} = m_{a,h} - r_{a,h} - r_{a,h-1} \quad (13)$$

copies of matching a in degree h . Such a minimal complex exists. Any homotopy equivalence between two minimal complexes is a chain isomorphism.

Proof. Because the scalar projection is an algebra map, $\bar{d}^2 = 0$. Therefore the right side of [equation \(13\)](#) is precisely the dimension of the residue homology in that matching and degree, and is nonnegative.

Existence follows either by successively cancelling unit entries, each of which removes two objects, or by the finite construction in [section 5](#). Cancellation is a chain-homotopy equivalence, and termination leaves a radical differential. Reduce any homotopy equivalence between the original complex and a minimal one modulo J . It remains a homotopy equivalence of vector-space complexes, while the minimal target has zero differential. Its term dimensions must equal the residue homology dimensions, proving the formula.

Now suppose f is a homotopy equivalence between two minimal complexes. Modulo J all differentials vanish, so a homotopy inverse is an actual inverse to $\bar{f}$ in every degree. The scalar matrix $\bar{f}$ is invertible within each matching block. Lift its inverse and factor f as its scalar invertible part times $1 + N$, where N is a matrix over J . The matrix ideal has the same nilpotence bound as J , since each entry of a long enough matrix product is a sum of vanishing products of radical elements. A finite geometric series inverts $1 + N$. Thus f is invertible in every degree; its inverse is a chain map because f was one. $\square$

4.1 Consequences for the existing cancellation strategy

For a fixed processed tangle, exact minimal multiplicities do not depend on the order of scalar-unit cancellations. Min-fill, LIFO, or another complete pivot strategy can change intermediate fill, coefficient density, and runtime, but not the final counts in [equation \(13\)](#). This is a stronger statement than agreement of final closed homology ranks: it applies before closure, matching by matching and degree by degree.

The formula is also a resource oracle. A caller can compute $R = \sum_{a,h} \mu_{a,h}$ before cobordism-valued Schur updates and decline an explicit representation whose minimal survivor count already exceeds a budget. That refusal is *not a knot verdict*. In particular, a large partial-tangle profile does not imply that its eventual closure is nontrivial.

The lower bound is representation-specific. It applies to finite direct sums of these matching objects up to chain homotopy. It does not rule out circuits, recurrences, macro complexes, compressed multiplicities, derived descriptions using different generators, or a decision procedure that avoids computing all Khovanov homology. This qualification is precisely where future quasi-polynomial progress may enter.

5 Finite transfer directly to the survivors

5.1 Constructing the scalar contraction

Let d_0 be the scalar lift of $\bar{d}$: equal-matching constant coefficients are multiplied by the corresponding identity cobordisms, and other entries are zero. These scalar blocks form a split subalgebra, so $d_0^2 = 0$. Put

$$\delta = d - d_0 = d + d_0.$$

Every entry of δ belongs to J .

Binary linear algebra decomposes each matching block of (C, d_0) as homology plus contractible two-term pairs. Let V denote the homology objects with the multiplicities [equation \(13\)](#). Choose maps

$$i : V \rightarrow C, \quad p : C \rightarrow V, \quad h : C^h \rightarrow C^{h-1}$$

whose entries are scalar identities and which satisfy

$$pi = 1, \quad d_0i = 0, \quad pd_0 = 0, \quad 1 + ip = d_0h + hd_0, \quad h^2 = hi = ph = 0. \quad (14)$$

One explicit construction chooses, in every degree, a basis B of boundaries, a complement H to B in the cycles, and lifts L of the next boundaries. In the basis (B, H, L) , let i include H , let p project to it, and let h map a boundary to its selected lift in the preceding degree. The implementation builds these bases with tracked binary column elimination.

Theorem 5.1 (Sharp finite transfer). *For $k \geq 1$, set*

$$I = \sum_{r=0}^{2k-1} (h\delta)^r i, \quad P = p \sum_{r=0}^{2k-1} (\delta h)^r, \quad (15)$$

$$H = \sum_{r=0}^{2k-1} (h\delta)^r h, \quad D = \sum_{r=0}^{2k-2} p\delta(h\delta)^r i. \quad (16)$$

Then (V, D) is minimal and the following identities hold:

$$\begin{aligned} D^2 = 0, \quad dI = ID, \quad Pd = DP, \quad PI = 1, \quad 1 + IP = dH + Hd, \\ H^2 = 0, \quad HI = 0, \quad PH = 0. \end{aligned} \quad (17)$$

Consequently it is homotopy equivalent to (C, d) . For $k = 0$, simply take $D = 0$ and the scalar contraction.

Proof. Matrices with entries in J form a nilpotent ideal of index at most $2k$, independently of matrix size. Since h is scalar, $h\delta$ and δh are in this ideal. Their geometric series therefore terminate as displayed. Each summand of D has at least one radical factor, proving minimality. The omitted possible last summand has $2k$ radical factors and vanishes.

For completeness, the algebra behind the perturbation step can be recorded without an infinite-series argument. Define

$$Q = \delta(1 + h\delta)^{-1} = (1 + \delta h)^{-1}\delta.$$

Finite multiplication gives

$$Q = \delta + \delta hQ = \delta + Qh\delta, \quad d_0Q + Qd_0 + QipQ = 0.$$

The last identity follows by substituting $ip = 1 + d_0h + hd_0$ and $d_0\delta + \delta d_0 + \delta^2 = 0$, then multiplying by the two finite inverses. In these terms,

$$D = pQi, \quad I = i + hQi, \quad P = p + pQh, \quad H = h + hQh.$$

For example, $D^2 = p(QipQ)i = p(d_0Q + Qd_0)i = 0$. Expanding $dI + ID$ and using the same three identities cancels all terms; the dual calculation gives $Pd + DP = 0$. Expansion of $dH + Hd$ gives $1 + IP$. The side conditions $h^2 = hi = ph = 0$ immediately imply $PI = 1$, $H^2 = 0$, $HI = 0$, and $PH = 0$ from the displayed formulas. These are the finite special-deformation-retract formulas of the classical perturbation lemma [4], here justified by the explicit radical bound. $\square$

5.2 Why the stated correction depth cannot be reduced

Proposition 5.2 (Sharp transfer order). *For each $k \geq 1$, there is a two-term complex with $4k - 2$ objects and exactly two residue survivors for which the first nonzero transferred-differential summand is*

$$p\delta(h\delta)^{2k-2}i.$$

Thus $2k - 2$ is the best possible universal correction-order bound for this category.

Proof. Take the $q = 2k - 1$ radical factors $f_1, \dots, f_q$ in the sharpness witness of [theorem 3.2](#), on a path of matching objects $a^{(0)}, \dots, a^{(q)}$. Place a survivor $U = a^{(0)}$ in degree zero and a survivor $V = a^{(q)}$ in degree one. For each internal path vertex add a contractible pair $B_j \xrightarrow{1} C_j$, with B_j in degree zero and C_j in degree one, both of matching type $a^{(j)}$.

Let d_0 consist of these identity arrows. Define δ by

$$U \xrightarrow{f_1} C_1, \quad B_1 \xrightarrow{f_2} C_2, \quad \dots, \quad B_{q-1} \xrightarrow{f_q} V.$$

All maps go from degree zero to degree one, so $d^2 = 0$ automatically. The scalar contraction sends C_j back to B_j . The only route from U to V in the transfer series uses all q radical factors and $q - 1$ copies of h . Its value is the nonzero fully dotted endomorphism. Every shorter route ends in a contractible target and is killed by p . The object count is $2 + 2(q - 1) = 4k - 2$. $\square$

This witness is an algebraic complex in the tangle category. We do not assert that every such witness is the scanner complex of a knot diagram. The theorem establishes sharpness for the algebraic transfer routine, not a topological worst-case lower bound for all recognizers.

5.3 A block-matrix pitfall

A scalar unit of $\text{End}(a)$ is an involution over $\mathbb{F}_2$, since every element with zero constant term squares to zero. This existing scalar shortcut must not be used for matrices. In $\mathbb{F}_2[x, y]/(x^2, y^2)$, let

$$N = \begin{pmatrix} 0 & x \\ y & 0 \end{pmatrix}.$$

Then $N^2 = xyI \neq 0$ and $N^3 = 0$, so

$$(I + N)^{-1} = I + N + N^2 \neq I + N.$$

This example is included in the executable tests. When a full block inverse is needed, geometric doubling is valid: powers of the *same* matrix commute, and

$$\prod_{j=0}^{s-1} (I + N^{2^j}) = \sum_{r=0}^{2^s-1} N^r.$$

Choose 2^s at least the nilpotence bound. No entrywise Frobenius rule for noncommuting matrix terms is being assumed.

5.4 Two-pass transfer using the marked splitting

Proposition 5.3 (One final square-zero correction). *Write a differential uniquely as*

$$d = d_B + ze, \quad d_B = d + z\partial d, \quad e = \partial d,$$

with entries of d_B, e in B_k . Then $d_B^2 = 0$ and $d_B e + e d_B = 0$. Suppose (I_B, P_B, H_B) contracts (C, d_B) to its minimal complex (V, D_B) . A minimal differential for d is

$$D = D_B + zP_B e I_B. \tag{18}$$

The corresponding contraction is

$$\begin{aligned} I &= I_B + zH_B e I_B, & P &= P_B + zP_B e H_B, \\ H &= H_B + zH_B e H_B. \end{aligned}$$

Proof. The direct sum decomposition $H_k = B_k \oplus zB_k$ and $z^2 = 0$ separate the two coefficients of d^2 : they are d_B^2 and $d_B e + e d_B$. Apply the perturbation lemma a second time, now with perturbation ze . Every product containing two such perturbations vanishes because z is central and $z^2 = 0$. Thus only the displayed correction remains. The scalar residues of d and d_B coincide, so their survivor objects coincide as well. Finally, D_B is radical and the extra term is divisible by z , hence the final differential is minimal. $\square$

The first pass may use the smaller radical index $2k - 1$ in B_k . A practical reduced-basis implementation would need only half the coefficients in each nonzero Hom space. Our supplied two-pass variant is instead a proof-oriented full-encoding implementation; its 308 complete contraction checks are reported below, but it is not included in the speedup claims.

6 Output-sensitive cost and the quasi-polynomial parameter regime

6.1 Computational model and input qualifications

Let M be the number of objects in a supplied complex before reduction, and let $R = \sum_{a,h} \mu_{a,h}$ be the exact survivor count. A morphism in a $2k$ -point category has at most 2^k coefficients. Let $\mathcal{A}(k)$ bound the bit cost of composing or adding two such morphisms, including the required topological-plan work. Standard finite basis arithmetic gives

$$\mathcal{A}(k) = \text{poly}(k) 2^{O(k)}.$$

This loose bound is enough for the asymptotic result. A compatible faster component-compression kernel can improve the exponential constant, but is not required for the proofs or the experiments here.

All complexity statements charge for packed bit strings, not merely for unit-cost arithmetic on arbitrarily large Python integers. Index and endpoint bookkeeping contributes logarithmic factors in the explicit input size and M ; we suppress these by $\tilde{O}$. A complex is assumed either to be known valid from its construction or to have been separately checked. Verifying $d^2 = 0$ by generic dense multiplication can itself cost $\tilde{O}(M^3\mathcal{A}(k))$. That cost must be added when requested.

Theorem 6.1 (Survivor-first construction cost). *On a valid input differential, a minimal differential and its inclusion map can be constructed in*

$$\tilde{O}\left(M^3 + M^2(1 + kR)\mathcal{A}(k)\right) \quad (19)$$

bit operations. The M^3 term is scalar binary linear algebra. The expensive cobordism-composition count in the transfer phase is $O((k+1)M^2R)$. A complete explicit contraction, including its $M \times M$ homotopy, can instead be produced and checked within

$$\tilde{O}\left(M^3 + (k+1)M^3\mathcal{A}(k)\right). \quad (20)$$

Proof. Build the scalar residue, compute its ranks, and construct the (B, H, L) bases from [section 5](#). Dense binary elimination and basis inversion cost $O(M^3)$ bit operations conservatively, after summing over the matching-degree blocks. Reading and forming the radical coefficient arrays is covered by $\tilde{O}(M^2\mathcal{A}(k))$.

To construct I , start with the $M \times R$ scalar matrix i and iterate

$$T_0 = i, \quad T_{r+1} = h(\delta T_r), \quad I = \sum_r T_r. \quad (21)$$

Do not form the $M \times M$ matrix $h\delta$ first. Each application of δ to T_r uses at most M^2R morphism compositions. Applying the scalar matrix h only sums coefficients, with no new surface composition. There are at most $2k - 1$ noninitial stages. Finally compute $D = p\delta I$. This proves [equation \(19\)](#); its extra input term also covers the case $R = 0$. When $k = 0$, the radical remainder is zero, so the construction is purely scalar.

For P , the analogous row iteration uses at most $O(kM^2R)$ compositions. Constructing H explicitly involves M columns rather than R , giving the $O(kM^3)$ bound. Each of the finitely many certificate identities can be checked by generic matrix multiplication in $O(M^3\mathcal{A}(k))$ time. This proves [equation \(20\)](#). $\square$

There are three distinct modes. The profile oracle performs only the binary rank calculation. The production transfer path can skip generic input-square verification when validity follows from the crossing construction. The proof-oriented path computes a full contraction and checks it. The code's `transfer(..., validate=False)` implements the second mode; its safer standalone default validates the input. The output-sensitive bound must not be advertised for that default without adding its validation cost.

Working coefficient matrices require $O(M^22^k)$ bits for the input and $O(MR2^k + R^22^k)$ bits for transfer columns and output, in addition to scalar basis data and index storage. This is an accounting of matrix payloads, not a tight bound on the shipped Python process: its geometry caches and object headers are additional. Discarding or bounding caches preserves the time bound. A conservative overall space bound is the stated time bound. A production memory claim should separately account for cache policy and representation metadata.

6.2 A comparison with scalar-unit cancellation

Generic dense cancellation can require $O((M - R)M^2)$ cobordism-valued Schur updates. The transfer construction replaces this coarse count by $O(kM^2R)$ expensive compositions for $k > 0$, plus binary work. The promising regime is therefore large cancellation, small R , and relatively expensive morphism arithmetic. There is no theorem saying it wins on every sparse instance: a good pivot sequence may make almost no fill, whereas scalar basis changes can be dense. An adaptive implementation should retain both paths.

If the radical remainder has minimum positive intrinsic weight $s > 1$, then a product of more than $\lfloor 2k/s \rfloor$ such factors vanishes already by weight. This yields a further safe truncation of [equation \(21\)](#); endpoint weights can refine it more. The supplied main implementation uses the uniform sharp bound and early termination on a zero iterate. Weight-aware truncation is an additional implementation opportunity, not part of the reported timings.

6.3 A complete parameterized recognition theorem

Let $N \geq 2$ be the explicit input size, including strands when a braid is supplied. Consider a *valid supplied arc-category scan*: every interface is modeled by some $\mathcal{C}_{k_t}$, and the crossing order and local attachments are specified, not found by an uncharged search. Braid scanning in [section 7](#) is one concrete instance. Define

$$k_* = \max_t k_t, \quad R_* = \max(1, \text{number of objects after minimal reduction at stage } t). \quad (22)$$

Lemma 6.2 (Constant local expansion). *Attaching one crossing to a circle-free matching creates at most two new closed circles in either smoothing. Consequently the number of objects immediately before the next elimination is at most eight times the preceding survivor count. For the braid attachment used in the harness, the bound is three times.*

Proof. Every new closed circle must contain at least one of the two newly attached smoothing arcs, since the old object has no closed circles. There are at most two such circles. Delooping c circles produces $2^c \leq 4$ copies in each of the two smoothings, for a total of at most eight. In a braid attachment the identity smoothing creates no circle, and the turnback smoothing creates at most one, giving $1 + 2 = 3$. $\square$

Theorem 6.3 (Parameterized recognition bound). *For a one-component diagram with a valid supplied scan as above, crossing attachment, survivor-first reduction, and exact closure give an exact unknot recognition algorithm. There is a fixed polynomial p for which its running time is bounded by*

$$T \leq p(N, k_*) 2^{O(k_*)} R_*^3. \quad (23)$$

The constants do not depend on the input. In particular,

$$k_* + \log_2(R_* + 1) = O(\log N) \quad \implies \quad T = N^{O(1)}, \quad (24)$$

$$k_* + \log_2(R_* + 1) = O((\log N)^2) \quad \implies \quad T = N^{O(\log N)}. \quad (25)$$

Proof. At each stage, [theorem 6.2](#) gives $M \leq 8R_*$. Local crossing maps can be assembled with $\text{poly}(N, k_*) 2^{O(k_*)} R_*^2$ work by finite-basis arithmetic. Apply [theorem 6.1](#), or even its full-certificate bound, at each of at most polynomially many elementary stages. Starting with the identity or

empty matching, the same expansion bound ensures $\log R_* = \text{poly}(N)$, so suppressed index costs are absorbed in p . The resulting cost is bounded as in [equation \(23\)](#).

A scan that ends with empty boundary already has a vector-space complex. For an open braid scan, closing a matching a by the fixed crossingless closure e yields $c(e, a) \leq k_*$ circles, so the explicit closed complex has at most $2^{k_*} R_*$ basis vectors. Binary Gaussian elimination costs at most the cube of this number, still within [equation \(23\)](#). Correctness follows from local chain-homotopy equivalence and the rank criterion proved in [theorem 7.1](#) below. Taking logarithms of the bound gives both parameter-regime consequences. $\square$

What this theorem does not charge for, and does not prove. It does not search over all crossing orders or supply a topological proof that an arbitrary knot admits a scan satisfying [equation \(25\)](#). The profile is an exact diagnostic once a stage differential has been constructed; it is not an independently available succinct certificate for the entire input. Thus [theorem 6.3](#) is an unconditional parameterized theorem and a conditional route to a *uniform* quasi-polynomial algorithm. The latter structural step remains open in this work.

7 An exact braid-scanning validation harness

The package includes a small complete algebraic harness to exercise the new reduction on genuine tangle complexes. It is deliberately separate from the optimized upstream `FastScan` representation.

Start with the identity matching between the top and bottom endpoints of a b -strand braid. For each braid generator attach its two-resolution complex at the relevant bottom endpoints, deloop any new circle, and reduce using either the transfer routine or an independent scalar-pivot baseline. Positive and negative generators use opposite choices of the two smoothings; the final homological degrees are shifted by the negative-crossing count. The geometry of the crossing and delooping plans is adapted from the inspected repository `geometry.py` and `algebra.py`. Static composition uses a previously archived numerical excerpt of `planar.py`.

For closure, use the representable functor

$$\text{Hom}(e, -) : \mathcal{C}_b \longrightarrow \text{finite-dimensional } \mathbb{F}_2\text{-vector spaces},$$

where e pairs each final bottom endpoint with its original top endpoint. A basis for $\text{Hom}(e, a)$ consists of dot monomials on the closed smoothing circles, and a differential $f : a \rightarrow a'$ acts by $g \mapsto fg$. This is the ordinary Frobenius evaluation of that crossingless closure. As an additive functor it preserves every chain-homotopy identity, so reducing the open complex before closing does not change the resulting homology.

Lemma 7.1 (The rank criterion used for knots). *For a knot K ,*

$$K \text{ is the unknot} \iff \dim_{\mathbb{F}_2} \text{Kh}(K; \mathbb{F}_2) = 2 \iff \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2) = 1.$$

Proof. Here is the characteristic-two relationship between the reduced and unreduced complexes. On each closed smoothing take the total dot derivative ∂ ; by the saddle calculation in [theorem 3.1](#) it commutes with the Khovanov differential. Let X be multiplication by a dot on a chosen marked component. It is also a chain map, $X^2 = 0$, and $\partial X + X \partial = 1$. Hence the unreduced complex splits into two isomorphic copies of $\ker \partial$, one multiplied by X , and the reduced complex can be identified with $\text{im } X = X \ker \partial$. For homological-degree counts, the two copies therefore have equal homology. Quantum shifts, which differ between the copies, are irrelevant here.

The unknot has reduced integral Khovanov homology $\mathbb{Z}$. Conversely, mod-two reduced rank one forces rational reduced rank at most one by the universal coefficient theorem. Rational reduced Khovanov homology of a knot has positive rank: its graded Euler characteristic is the normalized Jones polynomial, whose value at its normalization point is one. Its rational rank is therefore one. The unknot-detection theorem of Kronheimer–Mrowka then implies that K is the unknot [5]. $\square$

For links, the harness still computes unreduced homology, but the lemma is not used to make a knot verdict. A recognition wrapper must first check that the input has one component. For a braid closure this is the cycle count of the braid permutation. No resource limit or partial profile is substituted for that topological check or for a completed homology computation.

8 Reproducible computations and performance measurements

All reported calculations use exact integers and binary arithmetic. The Python standard library is sufficient; no floating-point arithmetic enters a theorem check, a rank, or a differential. Floating-point values below are wall-clock measurements only. Timings were obtained in this container with CPython 3.13.5; the full version and platform string, individual repetitions, and operation counts are recorded in `results/benchmark.json`.

8.1 Algebra, minimality, and homotopy certificates

The following checks were completed in this session.

Check	Count
Exhaustive basis-monomial products for $1 \leq k \leq 3$	2,244
Leibniz and homogeneous-weight checks on those products	2,244 each
Nonzero maximal radical words, $1 \leq k \leq 12$	12
Sharp-depth full transfer certificates, $1 \leq k \leq 8$	8
Random conjugated-complex full transfer certificates	500
Independent scalar-pivot minimal-profile comparisons	500
Marked two-pass full contraction certificates	308
Unit-test methods, including invalid input and matrix pitfall	18

The 500 random complexes start with a sum of contractible pairs and a minimal residual block, then undergo degree-preserving invertible elementary basis changes. Thus validity is built into their construction, not guessed from a random matrix. The verifier nevertheless checks the square of the differential, the square of the transferred differential, both chain-map identities, $PI = 1$, the homotopy identity, and all three side conditions. The last check confirms that the output differential has zero scalar residue.

For additional finite evidence, the script computes the entire radical-power filtration for $k \leq 4$ by linear spans of products. Here the first entry is $\dim H_k$, followed by $\dim J, \dim J^2, \dots$:

k	Dimensions of successive radical powers, including $J^0 = H_k$
1	2, 1, 0
2	12, 10, 6, 2, 0
3	104, 99, 82, 52, 22, 5, 0
4	1092, 1078, 1008, 826, 546, 266, 84, 14, 0

These data agree with the sharp index, but the proof of [theorem 3.2](#) is independent of finite testing. Similarly, the tested sharp transfer depths $0, 2, 4, \dots, 14$ are not the basis of the general sharpness proof.

8.2 Independent closed-braid comparisons

The braid test set consists of six explicit examples and 100 seeded random words for each of two, three, and four strands. Random words have between zero and nine crossings; the fixed seed is 202610072. The total is 306 closures, including both knots and links. For each case, both the local pivot scanner and the transfer scanner are compared degree by degree against an independent reduced crossing-cube implementation from the prior twist package. The expected unreduced profile is twice the reduced profile, by the chain splitting used in [theorem 7.1](#).

All comparisons passed. The transfer scanner also produced and verified a full contraction at every nonempty stage: 1,365 stage certificates in total. The two scanners agreed on every intermediate survivor count. Familiar calibration cases include the one-strand unknot with unreduced rank two, the closure of σ_1^3 with rank six, and the closure of $(\sigma_1\sigma_2^{-1})^2$ with rank ten.

The independence is limited and explicitly documented. The crossing-cube oracle assembles smoothing graphs and differential matrices independently and does not use the arc-transfer formulas. However, the static arc-composition excerpt and the crossing-plan conventions originate in the same project as the target scanner. We have not run the full upstream scanner, Rust port, or their complete regression suites in this environment.

8.3 Paired timings with favorable and unfavorable cases

Each timing is the median of five repetitions. Method order alternates, and geometry caches are fresh for each timed run. For fixed-complex tests, input construction and validation are outside the measured reduction. For braid scans, the two methods share the same local geometry harness; generic input-square validation and full certificate construction are disabled during timing for both. Exact output checks are performed separately. A ratio above one means transfer is faster than the included pivot baseline.

Fixed complex	M	R	Transfer (ms)	Pivot (ms)	Ratio
Gauge, $k = 3$	34	2	2.167	1.951	0.90
Gauge, $k = 3$	66	2	2.799	9.072	3.24
Gauge, $k = 4$	68	4	5.077	28.659	5.65
Gauge, $k = 3$	100	4	24.242	28.049	1.16
Sharp zigzag, $k = 8$	30	2	0.673	0.481	0.71

On these five fixed complexes the profile oracle alone took approximately 0.032–0.198 milliseconds. This is a particularly safe integration target: it is inexpensive relative to either full reduction path and gives an exact answer about minimal multiplicities.

Braid word	M	R	Transfer (ms)	Pivot (ms)	Ratio
σ_1^{25}	74	26	22.893	19.382	0.85
$(\sigma_1\sigma_2)^6$	39	17	18.751	14.028	0.75
$(\sigma_1\sigma_2^{-1})^6$	895	609	86.528	192.756	2.23
Braid-relator unknot	12	5	12.905	9.975	0.77

For the braid table, M and R are maxima across the scan. The last word is

$$(\sigma_1\sigma_2\sigma_1\sigma_2^{-1}\sigma_1^{-1}\sigma_2^{-1})^4\sigma_1\sigma_2.$$

The bracketed word is trivial by the braid relation, and the final closure is an unknot. Its test result is rank two. It is a useful warning against selecting only nontrivial knots or only dense algebraic matrices as performance evidence.

These measurements support an adaptive design, not unconditional replacement. The favorable alternating example has genuine tangle multiplicity growth; the sparse sharpness witness and several simple braids favor scalar cancellation. Moreover, an upstream recognition pipeline would reject many of the displayed nontrivial knots using cheaper filters, before reaching Khovanov homology. The measurements therefore do *not* establish a full-recognizer speedup or a speedup over the optimized upstream **FastScan**. Millisecond timings also have ordinary platform noise; all individual repetitions are retained so these ratios can be re-evaluated rather than treated as asymptotic evidence.

9 Integration strategy and correctness boundaries

The proposed destination is a new subtree such as `Topology/UnknotRecognition/research/radical_transfer/`. No current defaults should change solely on the basis of this report.

The lowest-risk first step is the non-mutating survivor oracle. The included adapter accepts the audited bit-valued reference **ScanComplex** interface, not the distinct optimized **FastScan** layout. It reads matching types, homological degrees, and equal-matching low bits. It neither composes cobordisms nor rewrites the upstream object graph. A stage instrumented before and after existing elimination should have exactly the predicted number of survivors. The adapter has a mock-interface test but has not been run against a complete upstream checkout; that remains an integration check.

The second step is an opt-in transfer backend with a tested conversion between matching IDs, circle-basis ordering, and sparse differential matrices. A full contraction verifier should be enabled on small development cases. The third step is paired profiling on inputs that actually reach the backend after the existing filters, including hard unknots and negative controls. Only those measurements can justify an adaptive production default.

Preserve the differential, not merely the counts. The profile formula does not say that the radical remainder can be discarded. An output with the correct objects and the zero differential will generally be wrong. The finite transfer terms are essential; [theorem 5.2](#) exhibits a surviving map that appears only at the maximum allowed order.

Preserve scope when interpreting a lower bound. Pivot-order invariance concerns the same processed tangle and the same category. Different scan orders give different prefixes. Twist compression and more succinct objects change the representation. Neither is contradicted by the minimality theorem.

Preserve budget and input semantics. A negative predicted multiplicity signals an invalid residue complex or an implementation error, not a knot property. A survivor ceiling produces a resource outcome, never “knotted.” A rank criterion is applied only to a completed calculation and

a verified one-component input. The simple Python harness has no certified production memory ceiling; its cache and allocation behavior must not be advertised as one.

Keep the verification claims at their actual level. Written proofs establish the stated algebraic theorems. Exact computations check finite instances and implementation identities. They are not a Lean formalization, an independent proof of every topological convention in a large codebase, or a certificate that the unrestricted asymptotic goal has been met. The archive includes seeds, raw timing repetitions, source identifiers, and a reproduction script to make further review concrete.

10 Further research questions and specific proof targets

Research question 10.1 (A cheaply enumerable family of favorable unknot models). Can one construct, in quasi-polynomial time, a quasi-polynomial-sized family of verified diagram transformations and scan orders such that every unknot input has at least one member satisfying [equation \(25\)](#)? The transformations must preserve the represented knot, and their encoding and execution costs must be included. Merely knowing that a favorable presentation exists is not enough.

This target can be weaker than controlling the profile of every knot and every scan. A precise sufficient condition is useful:

Proposition 10.2 (A one-sided model-cover criterion). *Suppose an algorithm constructs and verifies, in $N^{O(\log N)}$ time, at most $N^{O(\log N)}$ equivalent scan inputs, each of size at most $N^{O(\log N)}$. Suppose a fixed bound $T_0(N) = N^{O(\log N)}$ guarantees that, whenever the original input is the unknot, at least one of the resulting exact recognizer runs finishes within $T_0(N)$. Then bounded simulation of these runs decides unknot recognition in $N^{O(\log N)}$ time.*

Proof. Run each recognizer for $T_0(N)$ steps. Accept if any completed run identifies the unknot. Equivalent-input verification and exactness make acceptance sound. If none accepts, the stated covering guarantee rules out an unknot input. The product of the family-size bound and the per-run bound remains $N^{O(\log N)}$. Timed-out individual runs are not assigned verdicts; the negative conclusion uses the global covering theorem. $\square$

The proposition does not provide the required cover. Its purpose is to specify an exact topological/combinatorial target without incorrectly demanding that all nontrivial knots have small Khovanov complexes.

Research question 10.3 (Succinct minimal models beyond explicit multiplicities). Can the degree-indexed matching multiplicities and radical differentials be stored by circuits or recurrences whose size is quasi-polynomial even when R_* is exponential? A useful representation must support crossing attachment, composition, finite transfer, closure, and exact rank-one testing within proved bit costs. Integer-encoding of a large multiplicity alone does not provide the corresponding differential or a fast rank algorithm.

The three-braid distinction between exponentially large Betti numbers and polynomial computation in a more structured description [6] is a concrete reason to pursue this direction. The exact survivor theorem identifies what cannot be removed by ordinary pivoting, and therefore what must be represented more succinctly.

Research question 10.4 (A practical reduced-basis marked backend). Implement the algebra B_k in its $2^{c(a,b)-1}$ -dimensional Hom bases and combine it with [theorem 5.3](#). Can its compositions be evaluated directly through the earlier component kernel, without lifting to the full dot basis? Measure coefficient density and conversion costs, not just the nominal factor-two dimension reduction. Prove that the selected marked point is transported correctly through all local interfaces.

Research question 10.5 (A sharper support-dependent nilpotence bound). The universal correction order is sharp, but most measured complexes use only a small part of the arc algebra. Can one compute a smaller nilpotence bound for the ideal generated by the actual entries of δ , or for the operator $h\delta$ on the survivor columns, without doing the transfer first? Intrinsic weights give one inexpensive bound; radical-layer coordinates and forbidden matching transitions may improve it.

Research question 10.6 (Transfer-compatible twist and separator macros). Combine twist-normal-form complexes from the earlier package with this survivor-first reduction. What contracts must a macro expose for its inclusion, projection, and homotopy to remain compatible with subsequent gluing? For separator-based recursion, bound both boundary width and minimal multiplicity. A small number of geometric matching types by itself does not control the number of copies of those types.

Research question 10.7 (Adaptive reduction with a provable safety envelope). Can cheap predictors based on (M, R, k) , scalar rank density, and observed composition cost select transfer or scalar cancellation reliably? A full-run interleaving of two exact backends gives the usual constant-factor simulation bound relative to the faster run when preemption overhead is bounded. Local stage-by-stage switching is subtler: different minimal bases can alter the cost of future stages even when their object profiles coincide. Establish the appropriate claim rather than assuming local optimality implies global optimality.

Research question 10.8 (Proof-carrying profile and contraction certificates). Design serialized binary-rank and transfer certificates whose verification avoids trusting the elimination implementation. The scalar contraction identities already certify the predicted object counts, and [equation \(17\)](#) certifies the transferred complex. Can certificates be shared across many stages, with explicit bit-size bounds and without expanding all homotopy matrices? A compressed certificate requires a correspondingly proved verifier, not just a smaller file.

Research question 10.9 (Formalization of the characteristic-two core). Formalize the finite matching category, the dot derivation, weight additivity, and the sharp radical bound in Lean. A reusable second layer should prove finite homological perturbation in an additive category with a nilpotent ideal. The repository-specific interface can then instantiate these abstractions. Checking $d^2 = 0$ is necessary but does not establish that a generated complex represents the intended knot; local tangle equivalences remain part of the formalization target.

Research question 10.10 (Which abstract sharpness witnesses occur topologically?). The zigzag witnesses attain correction order $2k - 2$ in the algebraic category. Determine which can occur as subcomplexes or retracts of genuine knot-tangle complexes. A topological restriction excluding the worst algebraic witnesses could justify a better scanner-specific bound; a realizability construction would rule out that particular improvement. Neither conclusion follows from the finite algebra tests alone.

Research question 10.11 (A decision-specific interface invariant). Can a smaller invariant than the full minimal complex decide whether an arbitrary future closure has reduced Khovanov rank one, perhaps for a restricted family of closure contexts? Universal composition summaries already face

strong dimensional constraints. A decision-only summary may be smaller, but its closure domain and exactness must be stated, and positive and negative answers must both be justified.

Research question 10.12 (Separating algebraic and hierarchy complexity). The hierarchy program needs compressed cutting, controlled encodings, and quantitative bounds on its geometric routines. Can the present exact rank and nilpotence diagnostics serve as auxiliary verifiers or local cost estimates without replacing those missing topological arguments? Any synthesis should state a complete dependency graph and charge for every conversion between a hierarchy description and an algebraic tangle description.

11 Conclusion

The practical improvement is a change in the order of work: determine the exact minimal object profile with binary algebra first, then transfer the expensive radical differential directly to those survivors. The accompanying structural result is the sharp characteristic-two radical index $2k$, together with its marked splitting and the sharp transfer order $2k - 2$. These bounds are independent of matrix multiplicity; that is precisely why they are useful for large blocks of repeated matching objects.

The measurements support keeping both reduction strategies. They also support integrating the survivor oracle before making more invasive changes. The remaining global obstacle is no longer hidden inside an elimination heuristic: it is the need to control, avoid, or succinctly represent the exact minimal multiplicities and to find suitable presentations within a proved time bound. The report supplies a parameterized theorem and a concrete model-cover target, not an unrestricted quasi-polynomial recognizer.

A A six-object example where the last correction is essential

For $k = 2$, let $a = (01)(23)$ and $b = (03)(12)$. Let $f : a \rightarrow b$ and $g : b \rightarrow a$ be the undotted morphisms. Their composite is $gf = x_1 + x_2$ in $\text{End}(a)$. Take degree-zero objects (U, B_1, B_2) of types (a, b, a) and degree-one objects (C_1, C_2, V) of types (b, a, a) , with

$$d = \begin{pmatrix} f & 1 & 0 \\ 0 & g & 1 \\ 0 & 0 & x_1 \end{pmatrix}.$$

The scalar residue consists only of the two identity pivots. Its survivors are U and V . The degree-zero and degree-one labels are different copies even when they have the same matching type.

The direct term $p\delta i$ vanishes. The one-homotopy term $p\delta h\delta i$ also vanishes because it ends at C_2 . The last allowed term is

$$p\delta h\delta h\delta i = x_1 g f = x_1(x_1 + x_2) = x_1 x_2 \neq 0.$$

Thus replacing the original complex by two isolated objects would be wrong, even though it would have the correct minimal object profile. The exact transferred differential is the fully dotted map between them.

B Source audit and reproducibility map

The following Git object identifiers specify the inspected upstream versions. A hash identifies content; it is not a correctness certificate or an assertion that an entire checkout was tested.

Inspected source	Git object identifier
ProveIt fast/ subtree	b99a666269f660d932fc82ac8d1f29d3ae42231f
fastunknot/scan.py	4acff3cb5ffb4c4ad226a8bf6dcda3d8dc0bb561
fastunknot/algebra.py	d6d28604ac39c29145aa52f91074539a5aeac949
fastunknot/geometry.py	96acc160dbf2df7981b2db98b602311712a01531
fastunknot/planar.py excerpt provenance	1078526e7e7dbaf0b7267728105d870d94136769
openai/math binary-algebra section	29714bb7882fd989f987423540541957d8be6200

The archive is organized as follows.

Path	Role
article/article.tex, article.pdf	Complete source and compiled report
code/radical.py	Exact residue, contraction, transfer, marked variant, pivot baseline
code/braid_scan.py	Crossing-attachment and closure validation harness
vendor/reference_planar.py	Attributed static numerical composition excerpt
vendor/reference_cube.py	Independent reduced crossing-cube oracle from prior package
tests/	Unit tests and deterministic complex fixtures
experiments/	Algebra, braid, marked-variant, and paired timing scripts
results/	Completed-run data and logs
integration/	Non-mutating reference-interface adapter and deployment notes
PROVENANCE.json, CLAIMS.md	Source identity and exact claim boundaries
SHA256SUMS	Hash manifest of delivered files

The recommended reproduction commands are

```
python -m unittest discover -s tests -v
python experiments/validate_algebra.py
python experiments/validate_braids.py
python experiments/validate_marked.py
python experiments/benchmark.py
python experiments/make_tables.py
cd article
pdflatex -halt-on-error article.tex
pdflatex -halt-on-error article.tex
```

The benchmark script records fresh timings, which naturally need not reproduce identical milliseconds. The table generator keeps the numerical tables and summary macros consistent with the recorded benchmark file. The SHA-256 manifest covers the delivered snapshot; a local rerun changes outputs and therefore requires a new manifest to describe that new snapshot.

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- [10] ProveIt research contribution, *Twist-Compressed Khovanov Recognition: Explicit complexity bounds, exact cost certificates, and a tested route toward quasi-polynomial time*, 7 October 2026. Prior project artifact `unknot_twist_research_bundle.zip`; exact source artifact and hashes are recorded in `PROVENANCE.json`. Used for context and the independent crossing-cube validation oracle.
- [11] ProveIt research contribution, *Component Compression and Restart Entropy in Exact Unknot Recognition*, 7 October 2026. Prior project artifact `unknot_component_entropy_20261007.zip`; exact source artifact and hashes are recorded in `PROVENANCE.json`. Used for context and the attributed static composition excerpt.